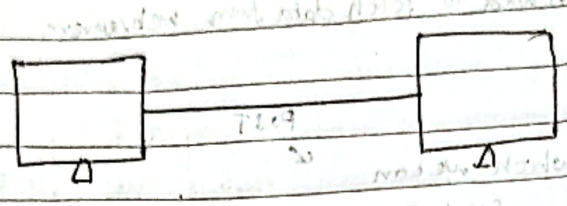
TCP & UDP



Whenever a computer wants to communicate with another computer the communication between those two computers needs to be a good and reliable

★ to gaurantee that data is recieved correctly

Ex If we want view a webpage we will try to view the file completely

& similarly if we want download a file we will download file complete rather than just a part of a file...

TCP

↳ Transmission control protocol

It is an main protocol used in TCP/IP Network.

★ TCP is an used to gaurantee that all data is recieved and in order,

without TCP some of the data could missing or out of order

Characteristics:-

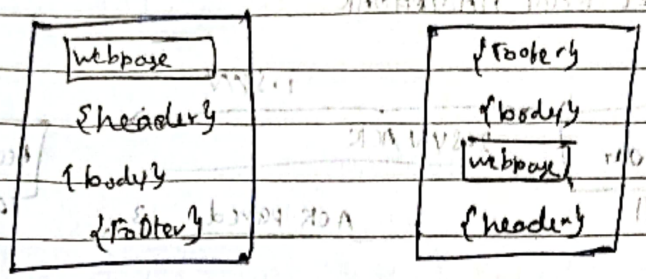
★ connection oriented
★ Reliable
★ ordered delivery
★ Error checking
★ Slower

used in:-

+ HTTP/HTTPS
+ File download
+ Banking apps

* The TCP will has an clear structured way which has an Proper way

Ex webpage



[No proper order will been presented in the way of data transmission].

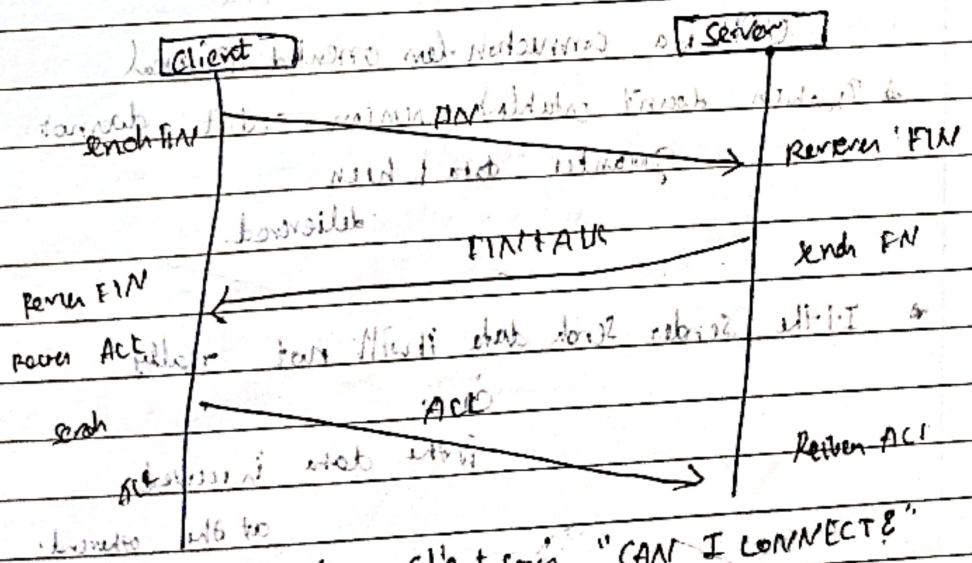
Image every thing will mixed & mixed up.

① Similary downloading a file

⇒ File will saved in a download in corrupted way that they wont clearly and or if..

TCP is an connection oriented version where it will clear "HAND SHAKE 3-way" for connecting

* Before sending a data it will first connect establishment



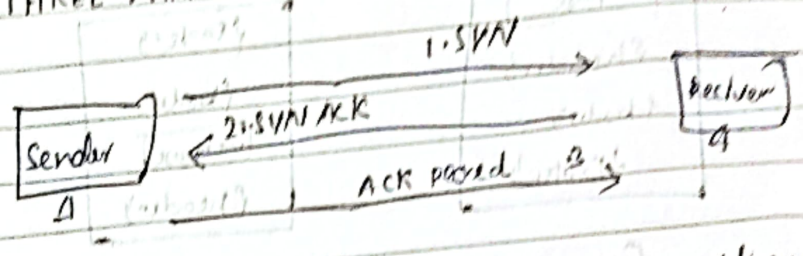
SYN - Client say "CAN I CONNECT?"

SYN-ACK - Server say "yes, I acknowledge"

ACK - client confirms

* Two computers will first establish the connection with each before transferring the data.

* THREE WAY HANDSHAK



* once it has been done the data will been delivered.

Another Important thing to remember about TCP, is that it guarantees the delivery of the data.

If anything goes wrong any data packet. It will resend them without any [No-Prob] it will send it again.

UDP

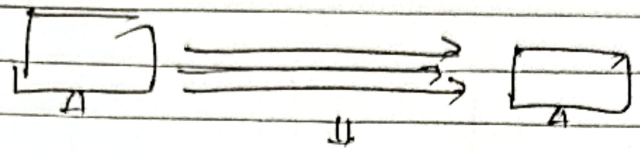
→ User Datagram Protocal:

UDP is also sending and recieving data.

* UDP is a connection-less oriented protocal
* It which doesn't establish a session and it doesnot garantee data has been delievered.

* If the sender send data it will not really care if the data is received at the other end.

* It won't get time syn/ACK before & during sending



It won't confirm reciever is getting the data or not..

UDP is known as "fire & forget protocol" It sends data it will not care what happened to data..

* And no data garuntee transmission can be held by the UDP

" UDP is faster than TCP "

Characteristics:

* Connectionless
* No garuntee of delivery
* Faster
* No ordering

Used in:

* video streaming
* online games
* voice calls (Zoom, whatsapp).